

DIGITAL PAYMENT SYSTEM

Objective / Aim

To design and implement a Digital Payment System in Java using an interface and custom exceptions, demonstrating wallet creation, adding money, UPI payments, balance checking, UPI validation, and transaction validation.

Problem Statement

Create a wallet with user name, mobile number, and UPI ID. Add money, make UPI payments, check balance, validate UPI IDs, prevent insufficient-balance payments, reject invalid amounts, handle exceptions, and display the final transaction status.

Algorithm / Logic

1. Create the PaymentService interface with pay() and checkBalance().
2. Create the Wallet class with private data members and a constructor.
3. Define addMoney(), getBalance(), and displayWalletDetails().
4. Create UPIPayment implementing PaymentService.
5. Create InsufficientBalanceException, InvalidUPIException, and InvalidAmountException.
6. Validate the UPI ID, amount, and available balance.
7. Throw the appropriate exception when validation fails.
8. Deduct the payment amount when all validations succeed.
9. Use try-catch-finally to handle the transaction.
10. Display the final balance and wallet details.

Steps for Execution

1. Create the Java project and add the classes/interface.
2. Compile and run DigitalPayment.java.
3. Enter the wallet details and initial balance.
4. Enter the amount to add.
5. Enter the receiver UPI ID and payment amount.
6. Observe the payment result and final wallet balance.

Source Code

```
import java.util.Scanner;

interface PaymentService {
    void pay(String upiId, double amount)
        throws InvalidUPIException, InvalidAmountException,
            InsufficientBalanceException;
    void checkBalance();
}

class InsufficientBalanceException extends Exception {
```

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        InsufficientBalanceException(String message) {
            super(message);
        }
    }

class InvalidUPIException extends Exception {
    InvalidUPIException(String message) {
        super(message);
    }
}

class InvalidAmountException extends Exception {
    InvalidAmountException(String message) {
        super(message);
    }
}

class Wallet {
    private String name;
    private String mobile;
    private String upiId;
    private double balance;

    Wallet(String name, String mobile, String upiId, double balance) {
        this.name = name;
        this.mobile = mobile;
        this.upiId = upiId;
        this.balance = balance;
    }

    void addMoney(double amount) {
        balance = balance + amount;
        System.out.println("Money added successfully.");
    }

    double getBalance() {
        return balance;
    }

    void reduceBalance(double amount) {
        balance = balance - amount;
    }

    void displayWalletDetails() {
        System.out.println("\n--- Wallet Details ---");
        System.out.println("Name      : " + name);
        System.out.println("Mobile   : " + mobile);
        System.out.println("UPI ID   : " + upiId);
        System.out.println("Balance  : " + balance);
    }
}

class UPIPayment implements PaymentService {
    private Wallet wallet;

    UPIPayment(Wallet wallet) {
        this.wallet = wallet;
    }

    public void pay(String upiId, double amount)
        throws InvalidUPIException, InvalidAmountException,
        InsufficientBalanceException {

        if (!upiId.contains("@"))
            throw new InvalidUPIException("Invalid UPI ID.");

        if (amount <= 0)
            throw new InvalidAmountException("Amount must be greater than zero.");

        if (amount > wallet.getBalance())
            throw new InsufficientBalanceException("Insufficient balance.");

        wallet.reduceBalance(amount);

        System.out.println("Payment successful.");
    }
}

```

```

        System.out.println("Amount Paid : " + amount);
        System.out.println("Paid To      : " + upiId);
    }

    public void checkBalance() {
        System.out.println("Current Balance : " + wallet.getBalance());
    }
}

public class DigitalPayment {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Mobile Number: ");
        String mobile = sc.nextLine();

        System.out.print("Enter UPI ID: ");
        String myUpi = sc.nextLine();

        System.out.print("Enter Initial Balance: ");
        double balance = sc.nextDouble();

        Wallet wallet = new Wallet(name, mobile, myUpi, balance);
        UPIPayment payment = new UPIPayment(wallet);

        System.out.print("Enter amount to add: ");
        double addAmount = sc.nextDouble();
        wallet.addMoney(addAmount);

        sc.nextLine();

        System.out.print("Enter Receiver UPI ID: ");
        String receiverUpi = sc.nextLine();

        System.out.print("Enter Payment Amount: ");
        double amount = sc.nextDouble();

        try {
            payment.pay(receiverUpi, amount);
        }
        catch (InvalidUPIException e) {
            System.out.println("Error: " + e.getMessage());
        }
        catch (InvalidAmountException e) {
            System.out.println("Error: " + e.getMessage());
        }
        catch (InsufficientBalanceException e) {
            System.out.println("Error: " + e.getMessage());
        }
        finally {
            System.out.println("Transaction process completed.");
        }

        payment.checkBalance();
        wallet.displayWalletDetails();

        sc.close();
    }
}

```